

NCERT Solutions For Class 12 Computer Science

Chapter 7 – Understanding Data

Exercise Answers

1. Identify data required to be maintained to perform the following services:

(a) Declare exam results and print e-certificates

Ans: Data required: Student name, roll number, class, exam scores/grades, subject/course information, year/session, certificate template.

(b) Register participants in an exhibition and issue biometric ID cards

Ans: Data required: Participant name, age, contact number, address, email, organization/institution detail, biometric data (like fingerprint or photo), ID card template.

(c) To search for an image by a search engine

Ans: Data required: Image name, tags/keywords, image file path or URL, image description, source metadata.

(d) To book an OPD appointment with a hospital in a specific department

Ans: Data required: Patient name, age, gender, contact number, address, previous medical history, preferred department, doctor, appointment date/time.

2. A school having 500 students wants to identify beneficiaries of the merit-cum means scholarship, achieving more than 75% for two consecutive years and having family income less than 5 lakh per annum. Briefly describe data processing steps to be taken by the school to prepare the list of beneficiaries.

Ans: The school should collect students' marks for the last two years and their family income data. Next, filter students who have scored above 75% in both years and have a family income less than 5 lakhs per annum. The filtered list represents the eligible scholarship beneficiaries.

3. A bank 'xyz' wants to know about its popularity among the residents of a city 'ABC' on the basis of number of bank accounts each family has and the average monthly account balance of each person. Briefly describe the steps to be taken for collecting data and what results can be checked through processing of the collected data.

Ans: Collect data from city households on number of bank accounts per family and average balance per person. Process the data to calculate total families holding accounts, total accounts per family, average monthly balances, and compare with other banks or total population. This analysis will reveal the bank's popularity in the city.

4. Identify type of data being collected/generated in the following scenarios:

(a) Recording a video – Ans: Unstructured data.

(b) Marking attendance by teacher – Ans: Structured data.

(c) Writing tweets – **Ans:** Unstructured data (may contain text, emojis, images etc.).

(d) Filling an application form online – **Ans:** Structured data.

5. Consider the temperature (in Celsius) of 7 days of a week as 34, 34, 27, 28, 27, 34, 34. Identify the appropriate statistical technique to be used to calculate the following:

(a) Find the average temperature.

Ans: Mean (average) is used to calculate average temperature.

Mean = $(34 + 34 + 27 + 28 + 27 + 34 + 34) / 7 = 218 / 7 = 31.14^{\circ}\text{C}$ (approx.).

(b) Find the temperature Range of that week.

Ans: Range = Maximum value – Minimum value = $34 - 27 = 7^{\circ}\text{C}$.

(c) Find the standard deviation temperature.

Ans: Standard deviation is the correct technique. Calculation: Mean = 31.14; squared deviations sum = 57.71; SD = $\sqrt{(57.71/7)} \approx 2.87^{\circ}\text{C}$.

6. A school teacher wants to analyse results. Identify the appropriate statistical technique to be used along with its justification for the following cases:

Ans: Compare performance in terms of division secured by students in Class XII A and Class XII B where each class strength is same.

Standard deviation should be used, as it measures variation/spread in the scores, providing a basis for comparison.

Comparison of class performance over five months (unit tests).

Mean should be used to calculate and compare the average performance over months.

7. Suppose annual day of your school is to be celebrated. The school has decided to felicitate those parents of the students studying in classes XI and XII, who are the alumni of the same school. In this context, answer:

(a) Which statistical technique should be used to find the number of students whose both parents are alumni?

Ans: Mode or counting technique should be used to find the frequency of such students.

(b) How varied are the age of parents of the students?

Ans: Standard deviation should be used to measure the variation in age.

8. For the annual day celebrations, the teacher is looking for an anchor in a class of 42 students. The teacher would make selection of an anchor on the basis of singing skill, writing skill, as well as monitoring skill.

(a) Which mode of data collection should be used?

Ans: Direct observation and evaluation or conducting auditions to assess skills.

(b) How would you represent the skill of students as data?

Ans: Represent each skill (singing, writing, monitoring) as columns in a

table and rate each student on a fixed scale (e.g., 1 to 5), creating structured data.

9. Differentiate between structured and unstructured data giving one example.

Structured data is organized in a fixed format, such as tables (e.g., attendance register, Excel sheet of marks). Unstructured data lacks a predefined structure, such as videos, audio files, emails, or free-form text documents.

The principal of a school wants to do following analysis on the basis of food items procured and sold in the canteen:

(a) Compare the purchase and sale price of fruit juice and biscuits.

(b) Compare sales of fruit juice, biscuits and samosa.

(c) Variation in sale price of fruit juices of different companies for same quantity (in ml).

Create an appropriate dataset for these items (fruit juice, biscuits, samosa) by listing their purchase price and sale price. Apply basic statistical techniques to make the comparisons.

Product	Company	Sales Price per Unit (₹)	Purchase Price per Unit (₹)	Units Sold
Fruit Juice (250 ml)	Real	40	35	10
Fruit Juice (250 ml)	Tropicana	45	30	15
Samosa	Samosa Party	30	27	20
Samosa	Chaayos	36	30	25
Samosa	Chai Point	33	29	15
Biscuit	Parle	10	6	30
Biscuit	Britannia	15	12	20
Biscuit	Sunfeast	20	17	15

Mean purchase price of fruit juice = $(35 + 30) / 2 = ₹32.5$

Mean sale price of fruit juice = $(40 + 45) / 2 = ₹42.5$

Profit per unit = ₹10

Mean purchase price of biscuits = $(6 + 12 + 17) / 3 = ₹11.67$

Mean sale price of biscuits = $(10 + 15 + 20) / 3 = ₹15$

Profit per unit = ₹3.33

Units sold: Biscuits = 65, Samosas = 60, Fruit juices = 25. Biscuits sold the most, followed by samosa and fruit juice.

Variation in sale price of fruit juices: Real (₹40) vs Tropicana (₹45) difference = ₹5 for same 250ml quantity.